

Database Management System

EXPERIMENT 4

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First, let's verify our tables have data:

```
-- Check EMPLOYEES table  
SELECT COUNT(*) FROM employees;
```

Expected Output:

```
COUNT(*)  
-----  
6
```

```
-- Check DEPARTMENT table  
SELECT COUNT(*) FROM department;
```

Expected Output:

```
COUNT(*)  
-----  
7
```

Basic SELECT Statements:

1. Show the structure of departments table and select all data

DESC department;

Expected Output:

Name	Null?	Type
DEPT_ID	NOT NULL	NUMBER(6)
DEPT_NAME	NOT NULL	VARCHAR2(20)
MANAGER_ID		NUMBER(6)
LOCATION_ID		NUMBER(4)

SELECT * FROM department;

Expected Output:

DEPT_ID	DEPT_NAME	MANAGER_ID	LOCATION_ID
10	Administration	200	1700
20	Marketing	201	1800
50	Shipping	124	1500
60	IT	103	1400
80	Sales	149	2500
90	Executive	100	1700
110	Accounting	205	1700

2. Show the structure of employees table and display specific columns

DESC employees;

Expected Output:

Name	Null?	Type
EMPLOYEE_ID	NOT NULL	NUMBER(6)

```

FIRST_NAME          VARCHAR2(20)
LAST_NAME    NOT    NULL  VARCHAR2(25)
EMAIL      NOT      NULL  VARCHAR2(25)
PHONE_NUMBER        VARCHAR2(20)
HIRE_DATE          NOT NULL DATE
JOB_ID            NOT NULL VARCHAR2(10)
SALARY            NUMBER(8,2)
COMMISSION_PCT      NUMBER(2,2)
MANAGER_ID          NUMBER(6)
DEPARTMENT_ID       NUMBER(4)
SELECT employee_id, last_name, job_id, hire_date
FROM employees
ORDER BY employee_id;

```

Expected Output:

EMPLOYEE_ID	LAST_NAME	JOB_ID	HIRE_DATE
100	King	AD_PRES	17-JUN-87
101	Kochhar	AD_VP	21-SEP-89
102	De Haan	AD_VP	13-JAN-93
103	Hunold	IT_PROG	03-JAN-90
104	Ernst	IT_PROG	21-MAY-91
107	Lorentz	IT_PROG	07-FEB-99

3. Provide an alias STARTDATE for the hire date

```

SELECT employee_id, last_name, job_id, hire_date AS
startdate
FROM employees;

```

Expected Output:

EMPLOYEE_ID	LAST_NAME	JOB_ID	STARTDATE
100	King	AD_PRES	17-JUN-87
101	Kochhar	AD_VP	21-SEP-89
102	De Haan	AD_VP	13-JAN-93
103	Hunold	IT_PROG	03-JUN-90
104	Ernst	IT_PROG	21-MAY-91
107	Lorentz	IT_PROG	07-FEB-99

4. Display unique job codes from the employee table

```
SELECT DISTINCT job_id FROM employees;
```

Expected Output:

```
JOB_ID
-----
AD_PRES
AD_VP
IT_PROG
```

5. Display last name concatenated with job ID

```
SELECT last_name || ', ' || job_id AS "Employee and Title"
FROM employees;
```

Expected Output:

```
Employee and Title
-----
King, AD_PRES
Kochhar, AD_VP
De Haan, AD_VP
```

Hunold, IT_PROG
Ernst, IT_PROG
Lorentz, IT_PROG

Advanced SELECT Features:

6. Display all data from employees with comma separation

```
SELECT employee_id || ', ' || first_name || ', ' ||  
last_name || ', ' ||  
email || ', ' || phone_number || ', ' || hire_date ||  
, ' ||  
job_id || ', ' || salary || ', ' || commission_pct ||  
, ' ||  
manager_id || ', ' || department_id AS "THE_OUTPUT"  
FROM employees;
```

Expected Output:

THE_OUTPUT

```
-----  
-----  
100, Steven, King, SKING, 515.123.4567, 17-JUN-87, AD_PRES,  
24000, , , 90  
101, Neena, Kochhar, NKOCHHAR, 515.123.4568, 21-SEP-89,  
AD_VP, 17000, , 100, 90  
102, Lex, De Haan, LDEHAAN, 515.123.4569, 13-JAN-93, AD_VP,  
17000, , 100, 90  
103, Alexander, Hunold, AHUNOLD, 590.423.4567, 03-JAN-90,  
IT_PROG, 9000, , 102, 60  
104, Bruce, Ernst, BERNST, 590.423.4568, 21-MAY-91, IT_PROG,  
6000, , 103, 60  
107, Diana, Lorentz, DLORENTZ, 590.423.4568, 07-FEB-99,
```

IT_PROG, 4200, , 103, 60

Arithmetic Operations:

7. Basic arithmetic operations with salaries

```
SELECT last_name, salary, salary + 300 AS "Salary+300"  
FROM employees;
```

Expected Output:

LAST_NAME	SALARY	Salary+300
King	24000	24300
Kochhar	17000	17300
De Haan	17000	17300
Hunold	9000	9300
Ernst	6000	6300
Lorentz	4200	4500

8. Complex arithmetic expressions

```
SELECT last_name, salary, 12 * salary + 100 AS "Annual1",  
       12 * (salary + 100) AS "Annual2"  
FROM employees;
```

Expected Output:

LAST_NAME	SALARY	Annual1	Annual2
King	24000	288100	289200
Kochhar	17000	204100	205200
De Haan	17000	204100	205200

Hunold	9000	108100	109200
Ernst	6000	72100	73200
Lorentz	4200	50500	51600

G. Working with commission percentages

```
SELECT last_name, job_id, salary, commission_pct,
       12 * salary * commission_pct AS "Annual Commission"
FROM employees
WHERE commission_pct IS NOT NULL;
```

Expected Output:

no rows selected

(Let's add some employees with commission for demonstration)

```
INSERT INTO employees VALUES
(149, 'Eleni', 'Zlotkey', 'EZLOTKEY', '011.44.1344.429018',
 TO_DATE('29-JAN-2000', 'DD-MON-YYYY'), 'SA_MAN', 10500,
 0.2, 100, 80);
```

```
INSERT INTO employees VALUES
(174, 'Ellen', 'Abel', 'EABEL', '011.44.1644.429267',
 TO_DATE('11-MAY-1996', 'DD-MON-YYYY'), 'SA_REP', 11000,
 0.3, 149, 80);
```

```
COMMIT;
```

Now run the commission query again:

```
SELECT last_name, job_id, salary, commission_pct,
       12 * salary * commission_pct AS "Annual Commission"
```

```
FROM employees
WHERE commission_pct IS NOT NULL;
```

Expected Output:

LAST_NAME		JOB_ID	SALARY
COMMISSION_PCT	Annual	Commission	
Zlotkey		SA_MAN	
10500	.2		25200
Abel		SA_REP	
11000	.3		39600

Column Aliases and Literals:

10. Using column aliases with and without AS keyword

```
SELECT last_name AS "Name", salary "Monthly Salary",
       salary * 12 "Yearly Salary"
FROM employees;
```

Expected Output:

Name	Monthly Salary	Yearly Salary
King	24000	288000
Kochhar	17000	204000
De Haan	17000	204000
Hunold	9000	108000
Ernst	6000	72000
Lorentz	4200	50400
Zlotkey	10500	126000

Abel	11000	132000
------	-------	--------

11. Using literal character strings

```
SELECT last_name || ' is a ' || job_id AS "Employee Details"
FROM employees;
```

Expected Output:

Employee Details

```
-----
King is a AD_PRES
Kochhar is a AD_VP
De Haan is a AD_VP
Hunold is a IT_PROG
Ernst is a IT_PROG
Lorentz is a IT_PROG
Zlotkey is a SA_MAN
Abel is a SA_REP
```

12. More complex literal example

```
SELECT last_name || '''s annual salary is $' ||
       TO_CHAR(salary * 12, '999,999') AS "Salary
Information"
FROM employees
WHERE ROWNUM <= 3;
```

Expected Output:

Salary Information

```
-----
King's annual salary is $288,000
```

Kochhar's annual salary is \$204,000
De Haan's annual salary is \$204,000

Working with NULL Values:

13. Handling NULL values in expressions

```
SELECT last_name, salary, commission_pct,  
       salary * (1 + NVL(commission_pct, 0)) AS "Total  
Compensation"  
FROM employees;
```

Expected Output:

LAST_NAME Compensation	SALARY	COMMISSION_PCT	Total
King	24000		
Kochhar	17000		
De Haan	17000		
Hunold	9000		
Ernst	6000		
Lorentz	4200		
Zlotkey	10500	.2	
Abel	11000	.3	

14300

Additional SELECT Features:

14. Using DUAL table for calculations

```
SELECT 15 * 25 AS "Calculation",  
       SYSDATE AS "Current Date",  
       USER AS "Current User"  
FROM DUAL;
```

Expected Output:

```
Calculation Current Da Current User  
-----  
375 23-JAN-24 SYSTEM
```

15. String manipulation with DUAL

```
SELECT 'Hello ' || 'World' AS "Greeting",  
       LENGTH('Oracle Database') AS "String Length",  
       UPPER('hello world') AS "Uppercase"  
FROM DUAL;
```

Expected Output:

```
Greeting      String Length UPPERCASE  
-----  
Hello World      14 HELLO WORLD
```

16. Display table structure using data dictionary

```
SELECT column_name, data_type, data_length, nullable
FROM user_tab_columns
WHERE table_name = 'EMPLOYEES'
ORDER BY column_id;
```

Expected Output:

COLUMN_NAME	DATA_TYPE	DATA_LENGTH	NULLABLE
-----	-----	-----	-----

EMPLOYEE_ID	NUMBER	6	N
FIRST_NAME	VARCHAR2	20	Y
LAST_NAME	VARCHAR2	25	N
EMAIL	VARCHAR2	25	N
PHONE_NUMBER	VARCHAR2	20	Y
HIRE_DATE	DATE	7	N
JOB_ID	VARCHAR2	10	N
SALARY	NUMBER	8	Y
COMMISSION_PCT	NUMBER	5	Y
MANAGER_ID	NUMBER	6	Y
DEPARTMENT_ID	NUMBER	4	Y

Final Comprehensive Example:

17. Complete employee information display

```
SELECT
    'Employee: ' || last_name || ', ' || first_name AS "Full
Name",
    'Job: ' || job_id AS "Position",
    'Hired: ' || TO_CHAR(hire_date, 'Month DD, YYYY') AS
"Start Date",
```

```

        'Salary: $' || TO_CHAR(salary, '999,999') AS "Monthly
Pay",
        'Department: ' || department_id AS "Dept ID"
FROM employees
ORDER BY last_name;

```

Expected Output:

Full Name	Position	Start
Date	Monthly Pay	Dept ID
Employee: Abel, Ellen	Job: SA_REP	May
11, 1996	Salary: \$11,000	Department: 80
Employee: De Haan, Lex	Job: AD_VP	January
13, 1993	Salary: \$17,000	Department: 90
Employee: Ernst, Bruce	Job: IT_PROG	May
21, 1991	Salary: \$ 6,000	Department: 60
Employee: Hunold, Alexander	Job: IT_PROG	January
03, 1990	Salary: \$ 9,000	Department: 60
Employee: King, Steven	Job: AD_PRES	June
17, 1987	Salary: \$24,000	Department: 90
Employee: Kochhar, Neena	Job: AD_VP	
September 21, 1989	Salary: \$17,000	Department: 90
Employee: Lorentz, Diana	Job: IT_PROG	February
07, 1999	Salary: \$ 4,200	Department: 60
Employee: Zlotkey, Eleni	Job: SA_MAN	January
29, 2000	Salary: \$10,500	Department: 80

Summary of SELECT Statement Capabilities:

1. **Projection** - Choosing specific columns
2. **Column Aliases** - Renaming column headings

3. **Arithmetic Operations** - Calculations in SELECT
4. **Concatenation** - Combining strings and columns
5. **Literal Values** - Including constant values
6. **DISTINCT** - Eliminating duplicate rows
7. **DUAL Table** - For calculations and functions
8. **NULL Handling** - Dealing with null values
9. **Formatting** - Using functions for better display